

1-D Source free, convection-diffusion problem using FINITE VOLUME METHOD (FVM)

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1 October 2026

Problem

Consider the transport of a general fluid property ϕ by convection and diffusion on a 1D domain of length $L = 1$ on the x axis $0 \leq x \leq 1$.

Conditions on the boundaries are Dirichlet boundary conditions :

1. $\phi = 1 = \phi_0$ at $x = 0$
2. $\phi = 0 = \phi_1$ at $x = 1$

Convection velocity is u and is uniform.

1. fluid property $\rho = 1 \text{ kgm}^{-3}$
2. Material property $\Gamma = 0.1 \text{ kg/m} \cdot \text{s}$

The Governing Equation :

$$\frac{d}{dx}(\rho u \phi) = \frac{d}{dx}(\Gamma \frac{d\phi}{dx}) \quad (1)$$

The exact solution to the problem is this problem is :

$$\frac{\phi - \phi_0}{\phi_L - \phi_0} = \frac{e^{\frac{\rho u x}{\Gamma}} - 1}{e^{\frac{\rho u L}{\Gamma}} - 1} \quad (2)$$

The Péclet number (Pe) compares the convective force to diffusive force.

- $Pe \ll 1$: diffusion dominates
- $Pe \approx 1$: convection and diffusion are comparable
- $Pe \gg 1$: convection dominates

We want to solve this 1D problem using FVM with Central difference scheme and we want to investigate how the péclet number governs the stability of the numerical scheme. To do so we first discretize the governing eqn using FVM and CD and treats the boundary nodes , write the equations in term of the péclet number. Then we,

1. Write a python code that solves for convection diffusion problem for $N = 50$, and $Pe = 1$.
 - Show that the central difference scheme predicts the exact solution quite well
2. Also **Run** the same code with $Pe = 5$ and observe what happens , explain why.

Discretization

Cell centered FVM is used to construct 1D grid of $N = 50$ Nodes/Control volumes. Grid spacing $h = \frac{L}{N}$
Integrate the governing equation (1) onto the finite volume centered at P.

$$\int \frac{d}{dx}(\rho u \phi) dV = \int \frac{d}{dx}(\Gamma \frac{d\phi}{dx}) dV \quad (3)$$

Using gauss divergence theorem :

$$\int \hat{n} \cdot (\rho u \phi) dA = \int \hat{n} (\Gamma \frac{d\phi}{dx}) dV \quad (4)$$

Resulting in the flux equation :

$$(\rho u \phi)_e - (\rho u \phi)_w = \Gamma A \left(\frac{d\phi}{dx} \right)_e - \Gamma A \left(\frac{d\phi}{dx} \right)_w \quad (5)$$

Given ρ cst., Γ cst , u cst. , h cst. , A cst.

We get :

$$\rho u (\phi)_e - (\phi)_w = \Gamma \left(\frac{d\phi}{dx} \right)_e - \left(\frac{d\phi}{dx} \right)_w \quad (6)$$

Let's assume that ϕ varies piecewise linear between nodes , we choose Central differencing.

- $(\phi)_e = \frac{\phi_E + \phi_P}{2}$
- $(\phi)_w = \frac{\phi_W + \phi_P}{2}$
- $\left(\frac{d\phi}{dx} \right)_e = \frac{\phi_E - \phi_P}{h}$
- $\left(\frac{d\phi}{dx} \right)_w = \frac{\phi_P - \phi_W}{h}$

Replacing into EQN (6):

$$\frac{\rho u}{\Gamma} \left(\frac{\phi_E + \phi_P}{2} - \frac{\phi_W + \phi_P}{2} \right) = \frac{\phi_E - \phi_P}{h} - \frac{\phi_P - \phi_W}{h} \quad (7)$$

Using that the cell péclet number $Pe = \frac{\rho u h}{\Gamma}$

$$\frac{Pe}{2} (\phi_E - \phi_W) = \phi_E - 2\phi_P + \phi_W \quad (8)$$

$$2\phi_P = \left(1 - \frac{Pe}{2}\right)\phi_E + \left(1 + \frac{Pe}{2}\right)\phi_W \quad (9)$$

- $a_P = a_W + a_E = 2$
- $a_W = \left(1 + \frac{Pe}{2}\right)$
- $a_E = \left(1 - \frac{Pe}{2}\right)$

FOR ALL INTERIOR NODES

FOR NODE 1 : Boundary node , $\phi_0 = 1$ at $x = 0$

Using the flux eqn. (6) :

- $\phi)_e = \frac{\phi_E + \phi_P}{2}$, $\phi_w = \phi_0 = 1$
- $\frac{d\phi}{dx})_e = \frac{\phi_E - \phi_P}{h}$, $\frac{d\phi}{dx})_w = \frac{\phi_P - \phi_0}{\frac{h}{2}} = \frac{2(\phi_P - 1)}{h}$

$$\frac{\rho u}{\Gamma} \left(\frac{\phi_E + \phi_P}{2} - 1 \right) = \frac{\phi_E - \phi_P}{h} - 2 \left(\frac{\phi_P - 1}{h} \right) \quad (10)$$

$$Pe \left(\frac{\phi_E}{2} - 1 + \frac{\phi_P}{2} \right) = \phi_E - 3\phi_P + 2 \quad (11)$$

$$\left(3 + \frac{Pe}{2}\right)\phi_P = \left(1 - \frac{Pe}{2}\right)\phi_E + (2 + Pe) \quad (12)$$

- $a_P \phi_P = a_E \phi_E + (2 + Pe)$
- $a_E = 1 - \frac{Pe}{2}$

FOR NODE 5 : Boundary node , $\phi_L = 0$ at $x = L$

Using the flux eqn. (6) :

- $\phi)_e = \phi_L = 0$, $\phi)_w = \frac{\phi_W + \phi_P}{2}$
- $\frac{d\phi}{dx})_e = \frac{\phi_L - \phi_P}{\frac{h}{2}}$, $\frac{d\phi}{dx})_w = \frac{\phi_P - \phi_W}{h}$

$$\begin{aligned}
\frac{\rho u}{\Gamma}[-(\frac{\phi_P + \phi_W}{2})] &= [(-\frac{2\phi_P}{h}) - (\frac{\phi_P - \phi_W}{h})] \\
Pe[-(\frac{\phi_P + \phi_W}{2})] &= -3\phi_P + \phi_W \\
(3 - \frac{Pe}{2})\phi_P &= (1 + \frac{Pe}{2})\phi_W \\
a_P\phi_P &= a_W\phi_W \\
a_P = a_W + (2 - Pe) &= (3 - \frac{Pe}{2}), a_W = 1 + \frac{Pe}{2}
\end{aligned} \tag{13}$$

VALID FOR LAST NODE

Numerical implementation

- For interior nodes : $\phi_{new}[i] = (a_E\phi[i+1] + a_W\phi[i-1])/a_P$
- For Node 1 : $\phi_{new}[i] = (a_E\phi[i+1] + (2 + Pe))/a_P$
- For last Node : $\phi_{new}[i] = (a_W\phi[i-1])/a_P$

Numerical Results

For $N = 50$ and $Pe = 1$:

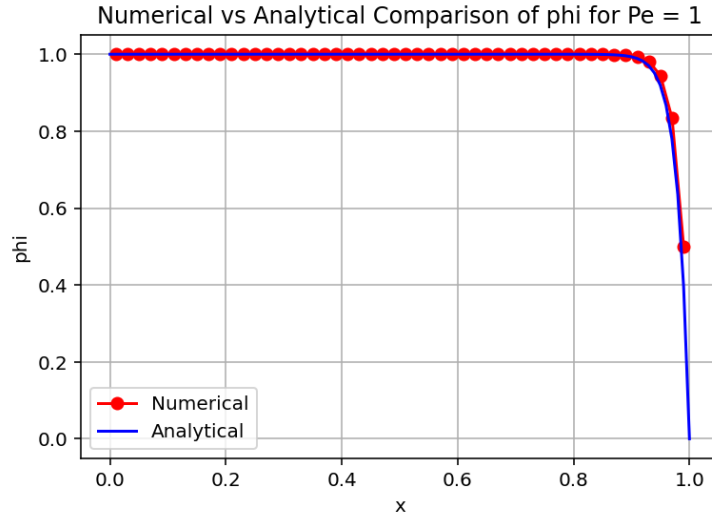


Figure 1: Plot of Numerical/analytical value of phi Vs x for $Pe = 1$

Table illustrates a comparison between Numerical and Analytical value of phi for $Pe = 1$

Node	Distance	Numerical solution of phi	Analytical solution of phi
1	0.01	1.0	1.0
2	0.03	1.0	1.0
3	0.05	1.0	1.0
4	0.07	1.0	1.0
5	0.09	1.0	1.0
6	0.11	1.0	1.0
7	0.13	1.0	1.0
8	0.15	1.0	1.0
9	0.17	1.0	1.0
10	0.19	0.9999999999999999	1.0
11	0.21	0.9999999999999998	1.0
12	0.23	0.9999999999999993	1.0
13	0.25	0.9999999999999999	0.9999999999999999
14	0.27	0.9999999999999977	0.9999999999999999
15	0.29	0.9999999999999964	0.9999999999999997
16	0.31	0.9999999999999929	0.9999999999999989
17	0.33	0.9999999999999892	0.9999999999999971
18	0.35000000000000003	0.9999999999999784	0.9999999999999921
19	0.37	0.9999999999999677	0.9999999999999785
20	0.39	0.9999999999999357	0.9999999999999415
21	0.41000000000000003	0.9999999999999027	0.99999999999998411
22	0.43	0.9999999999998035	0.99999999999995677
23	0.45	0.999999999999683	0.9999999999998245
24	0.47000000000000003	0.9999999999993219	0.99999999999968032
25	0.49	0.9999999999987248	0.99999999999913075
26	0.51	0.9999999999969333	0.99999999999763666
27	0.53	0.9999999999928629	0.99999999999357504
28	0.55	0.9999999999806515	0.99999999998253498
29	0.57000000000000001	0.9999999999474364	0.9999999999525296
30	0.59	0.99999999998477911	0.999999999987098761
31	0.61	0.99999999995576306	0.999999999964941507
32	0.63	0.999999999986871494	0.999999999904740661
33	0.65	0.99999999960977615	0.999999999741194591
34	0.67	0.99999999883295976	0.999999999296944586
35	0.69000000000000001	0.99999999650793931	0.9999999998090344152
36	0.71	0.99999998953287799	0.999999994813565594
37	0.73	0.99999996862077785	0.999999985915878759
38	0.75	0.99999990588447745	0.99999961758359449
39	0.77	0.9999971770642034	0.9999896178346452
40	0.79	0.99999153172249	0.9999718172892246
41	0.81	0.9999745964066755	0.9999235073570658
42	0.83000000000000001	0.9999237904592324	0.9997924143687108
43	0.85	0.9997713741984959	0.9994367331127918
44	0.87	0.9993141254162865	0.9984718393872984
45	0.89	0.9979423824412507	0.9958546603809587
46	0.91	0.9938271535161428	0.9887569019810261
47	0.93	0.9814814733796237	0.9695108714043179
48	0.95000000000000001	0.9444444329700664	0.9173323729010483
49	0.97	0.8333333226228747	0.7758928976437464
50	0.99	0.49999999158129943	0.3925596548440143

Table 1: My data for Numerical value of phi for $Pe = 1$

For $N = 50$ and $Pe = 5$:

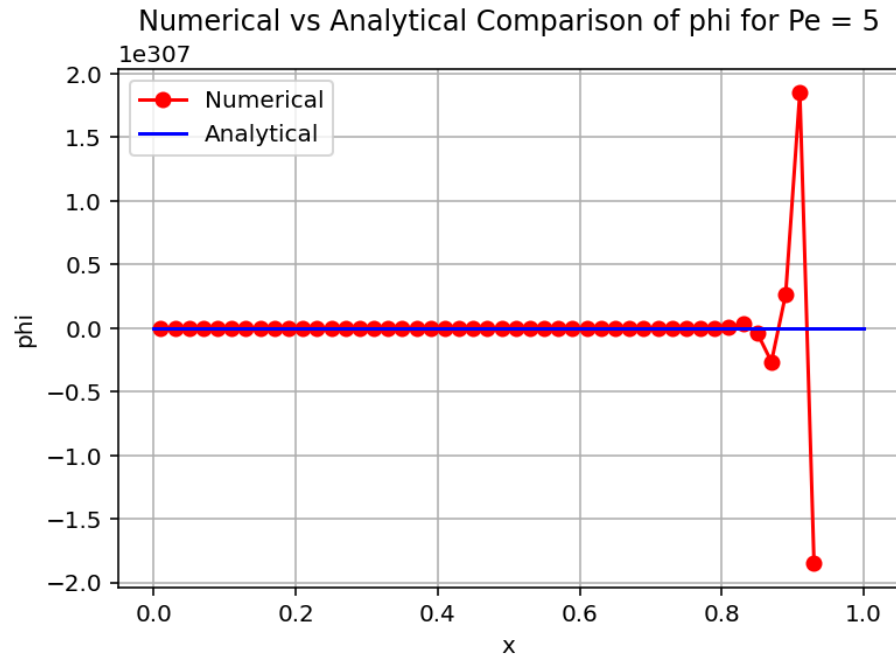


Figure 2: Plot of Numerical/analytical value of phi Vs x for $Pe = 5$

Table illustrates a comparison between Numerical and Analytical value of phi for $Pe = 1$

Node	Distance	Numerical solution of phi	Analytical solution of phi
1	0.01	1.3528263849153498e+287	1.0
2	0.03	3.4722543879493975e+288	1.0
3	0.05	-4.3140130274522824e+288	1.0
4	0.07	-3.2162194684339374e+289	1.0
5	0.09	3.2816895848397174e+289	1.0
6	0.11	2.3124590698824843e+290	1.0
7	0.13	-2.3175511900473782e+290	1.0
8	0.15	-1.6234739944049735e+291	1.0
9	0.17	1.6238700481955765e+291	1.0
10	0.19	1.1368014462880442e+292	1.0
11	0.21	-1.1368322504717578e+292	1.0
12	0.23	-7.957897629730968e+292	1.0
13	0.25	7.957921588540523e+292	1.0
14	0.27	5.570550702367262e+293	1.0
15	0.29	-5.570552565830227e+293	1.0
16	0.31	-3.8993872308891843e+294	1.0
17	0.33	3.8993873758251925e+294	1.0
18	0.35000000000000003	2.729571196896036e+295	1.0
19	0.37	-2.7295712081688365e+295	1.0
20	0.39	-1.9106998483485064e+296	1.0
21	0.41000000000000003	1.9106998492252796e+296	1.0
22	0.43	1.3374898946622763e+297	1.0
23	0.45	-1.33748989473047e+297	1.0
24	0.47000000000000003	-9.362429263272407e+297	1.0
25	0.49	9.362429263325447e+297	1.0
26	0.51	6.5537004843401895e+298	1.0
27	0.53	-6.553700484344315e+298	1.0
28	0.55	-4.587590339041982e+299	1.0
29	0.57000000000000001	4.587590339042303e+299	1.0
30	0.59	3.211313237329687e+300	1.0
31	0.61	-3.2113132373297117e+300	1.0
32	0.63	-2.247919266130804e+301	1.0
33	0.65	2.247919266130806e+301	1.0
34	0.67	1.5735434862915647e+302	1.0
35	0.69000000000000001	-1.5735434862915649e+302	1.0
36	0.71	-1.1014804404040954e+303	1.0
37	0.73	1.1014804404040954e+303	1.0
38	0.75	7.710363082828668e+303	1.0
39	0.77	-7.710363082828668e+303	1.0
40	0.79	-5.397254157980068e+304	1.0
41	0.81	5.397254157980068e+304	1.0
42	0.83000000000000001	3.778077910586048e+305	1.0
43	0.85	-3.778077910586048e+305	0.9999999999999999
44	0.87	-2.6446545374102335e+306	0.9999999999999862
45	0.89	2.6446545374102335e+306	0.99999999999980447
46	0.91	1.8512581761871635e+307	0.9999999997257663
47	0.93	-1.8512581761871635e+307	0.9999999620288001
48	0.95000000000000001	-inf	0.9999948277163905
49	0.97	inf	0.999310467197038
50	0.99	inf	0.9107622823924667

Table 2: My data for Numerical value of phi for $Pe = 5$

Discussion of Results

Consider the Global Péclet number $Pe_L = \frac{PeL}{h}$ For case (a) i.e $Pe = 1$ we have a $Pe_L = 50$ and for case (b) i.e $Pe = 5$ we have a $Pe_L = 250$. The two cases present a global péclet number which is $\gg 1$ which shows that convection force is way stronger than diffusion force over the whole domain. Physically we expect the property ϕ (initial value upstream) to be transported over a large distance before diffusion significantly changes it, convection transports ϕ faster than diffusion can spread it.

NOTE : CONVECTION carries/draggs the value , Diffusion spreads the value

Diffusion starts to spread the value of ϕ after $x = 0.8$ for both cases.

Figure 2 shows that the numerical scheme produces a solution which is not bounded , therefore we can say that the scheme does not respect boundedness.

Boundedness says that in the absence of a source , the value of ϕ at internal nodes should be bounded by min and max of boundary values i.e 0 and 1 which is clearly not the case because :

1. ϕ at **Node 1** = 1.3528263849153498e+287 $\gg 1$
2. ϕ at **Node 18** = 2.729571196896036e+295 $\gg 1$
3. ϕ at **Node 20** = -1.9106998483485064e+296 $\ll 0$

This is due to the coefficients involved in the discretisation scheme being of different signs. By convention we want all the coefficients to be positive.

For interior nodes we have

- $a_W = 1 + \frac{Pe}{2}$
- $a_E = 1 - \frac{Pe}{2}$

Clearly to have a_W and $a_E \geq 0$ we require $-2 \leq Pe \leq 2$, which is violated for case (b) $Pe = 5$. This generates an unbounded solution.

For case (a) the grid péclet number $Pe = 1$ which lies $-2 \leq Pe \leq 2$ predicts that diffusion force and convection force are equally comparable across a mesh which fits perfectly to the results because Central difference scheme produces an accurate result and a bounded solution.

1. max value ϕ at **Node 1** = 1.0 ≤ 1
2. min value of ϕ at **Node 50** = 0.49999999158129943 > 0

Conclusion

From this investigation we can say that Central difference scheme in Finite Volume Method produces accurate results for $-2 \leq Pe \leq 2$. Beyond the Pe value of 2 the results get infinitely large/small (wiggles !!) around the exact solution.